ECE303 Receivers, Antennas, and Signals

Electronics and Communication Engineering

University of Kufa

2019

Introduction

ECE303 Receivers, Antennas, & Signals

Outline

- Basics.
- Types of Antenna.
- Radiation Mechanism.
- Formation and Detachment of E lines.
- .Current Distribution.
- Current Variation.

Constantine A. Balanis, Antenna Theory, 4thEd., Wiley, 2016.

Basics

What is an Antenna

Merriam-Webster:

A usually metallic device (such as a rod or wire) for radiating and receiving radio waves.

IEEE:

The part of a transmitting or receiving system that is designed to radiate or to receive electromagnetic waves.

Constantine A. Balanis:

The transitional structure between free-space and a guiding device.

What do Antennas Do?

- Convert between guided wave and external propagating wave
- Configure the radiation pattern
- State the polarization
- Connect with other antennas

Antenna As A Transition/Transducer Device

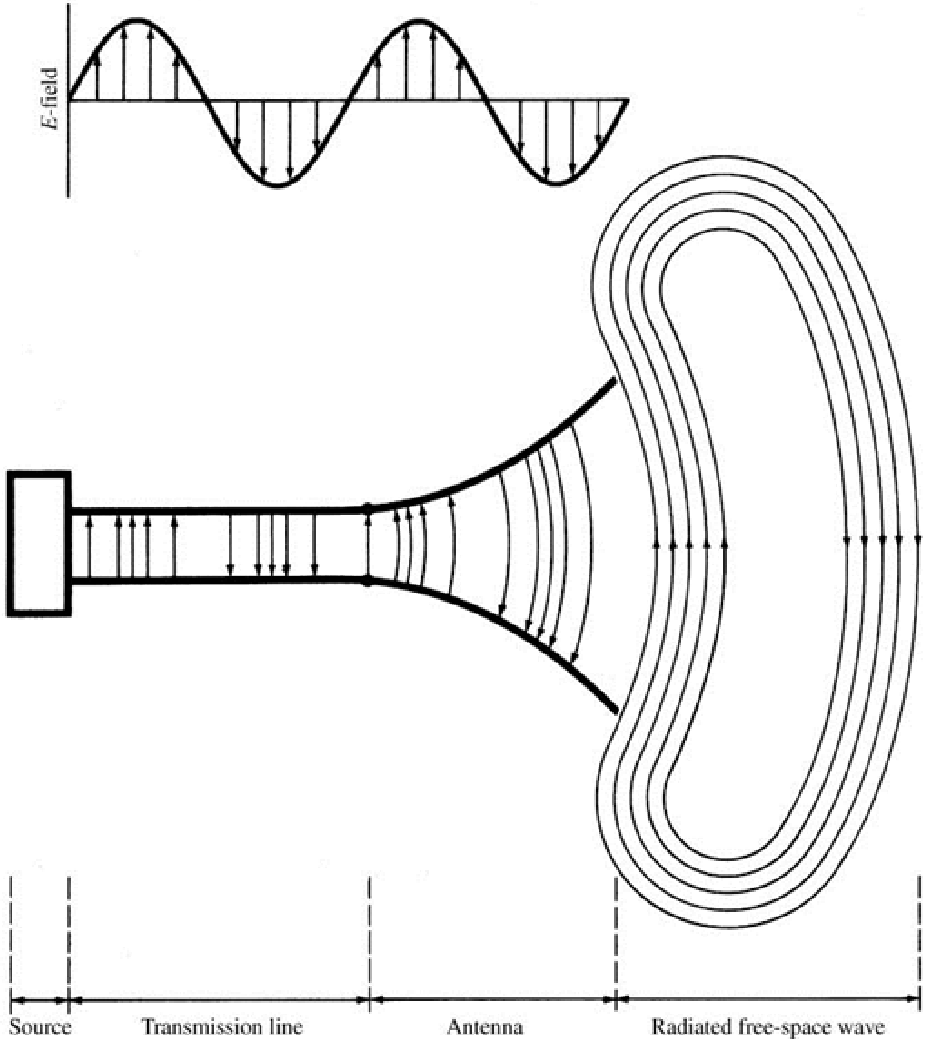


Fig. 1.1

Chapter 1
Antennas

Thevenin Equivalent
In Transmission Mode

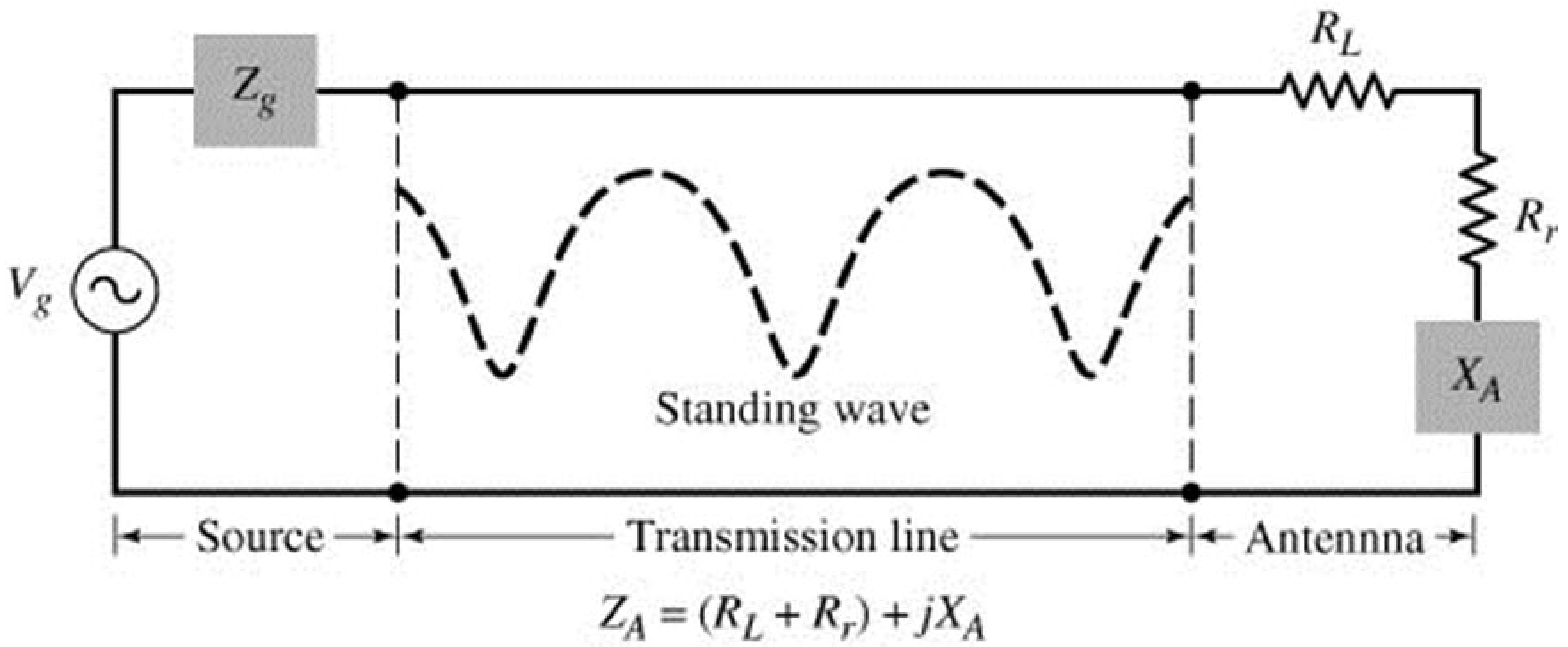


Fig. 1.2

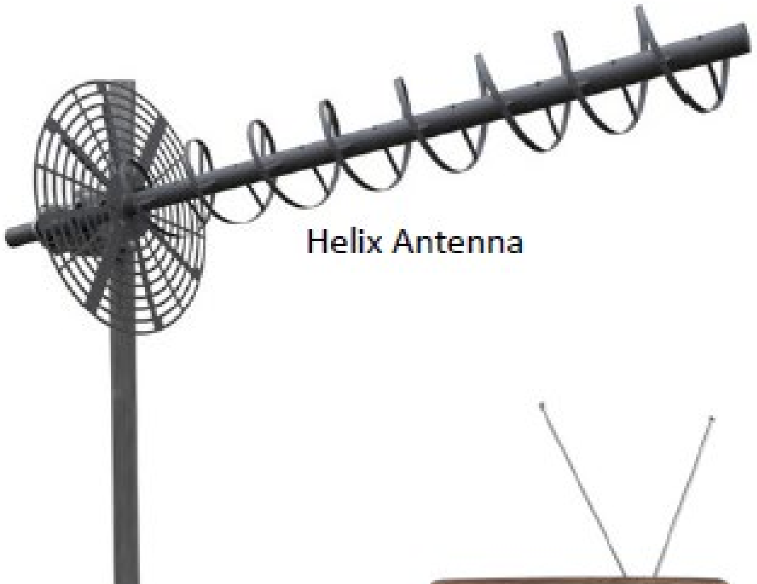
Chapter 1
Antennas

Types of Antenna

Antenna Categories

- Wire Antenna.
- Aperture Antenna.
- Microstrip Antenna.
- Array Antenna.
- Reflector Antenna.
- Lens Antenna.
- Other Antenna for Mobile Units.

Wire Antenna



Helix Antenna

Dipole Antenna

Loop Antenna

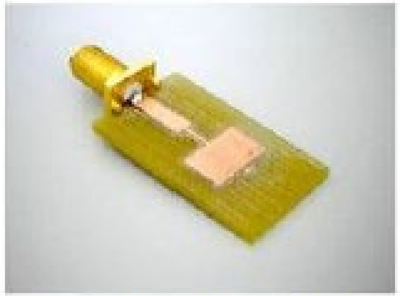
Aperture Antenna

Pyramidal Horn

Rectangular Waveguide

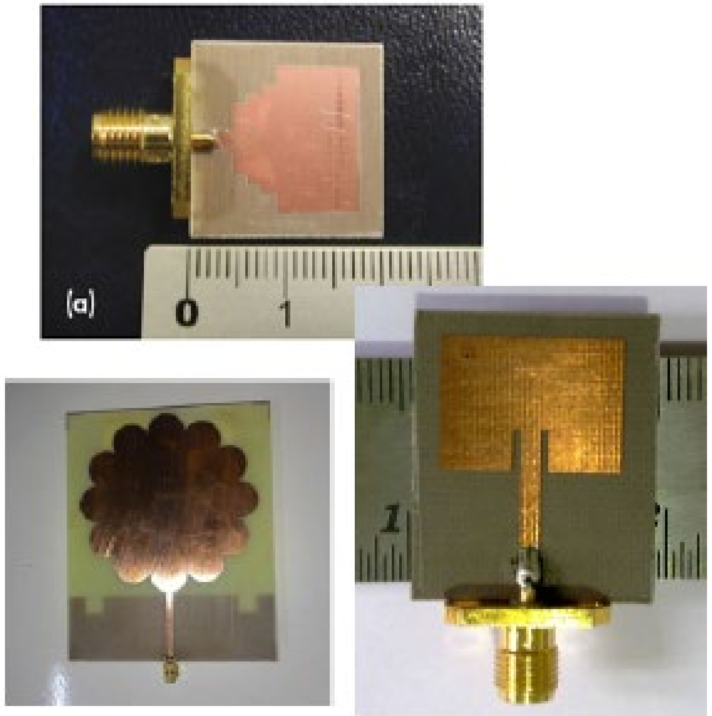
Conical Horn

Microstrip Antenna



Rectangular Patch

Circular Patch

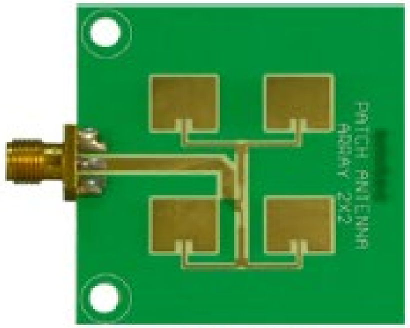


Other Variants

Array Antenna


Yagi-Uda

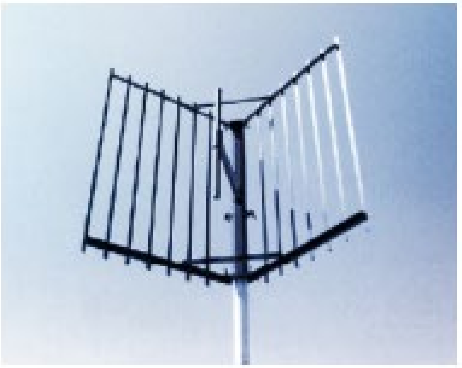

Phased Array



Patch Array


Slotted Waveguide

Reflector Antenna



Reflector Grid

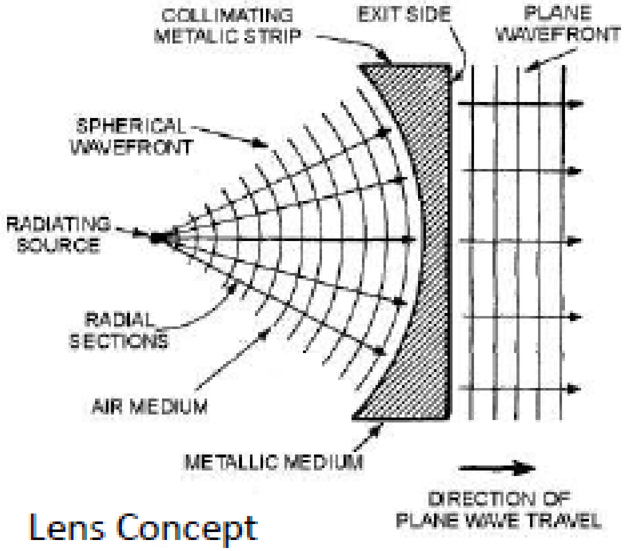
Parabolic Dishes

Corner Reflector

Lens Antenna

Conical Horns + Lens

Ball Lens



Lens Concept

Radiation Mechanism

Charge Uniformly Distributed in a Circular Cross Section Cylinder

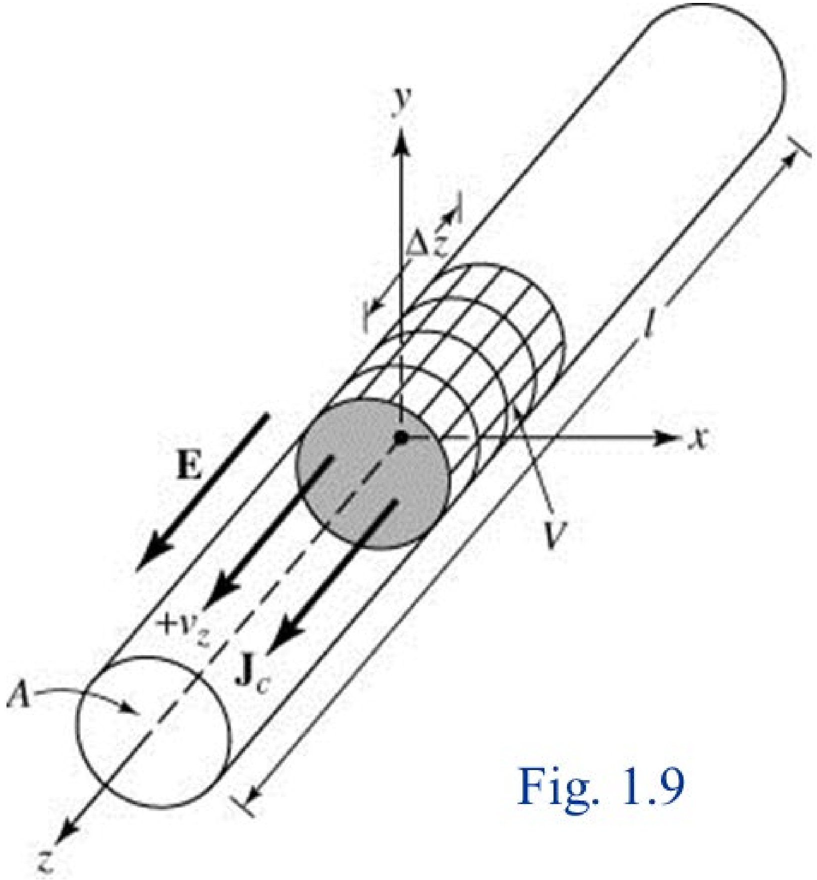


Fig. 1.9

Chapter 1
Antennas

Single Wire

$$J_z = q_v v_z \qquad (1\text{-}1a)$$

$$J_s = q_s v_z \qquad (1\text{-}1b)$$

$$I_z = q_\ell v_z \qquad (1\text{-}1c)$$

$$\frac{dI_z}{dt} = q_\ell \frac{dv_z}{dt} = q_\ell a_z \qquad (1\text{-}2)$$

$$\boxed{\ell \frac{dI_z}{dt} = \ell q_\ell \frac{dv_z}{dt} = \ell q_\ell a_z} \qquad (1\text{-}3)$$

Since $qa = q\dot{v}$ produces radiation, it follows that *dI/dt* produces radiation.

a. Use charges for transients and pulses
b. Use currents for time-harmonic variations

Therefore:

To create radiation, there must be:

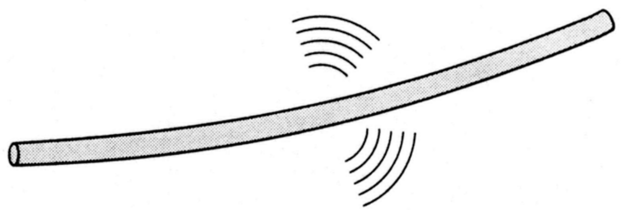
a. Time-varying current

or

b. Acceleration (or deceleration) of charge

1. If a charge is not moving, current is not created and there is no radiation.
2. If charge is moving with a uniform velocity:
   a. There is no radiation if the wire is straight, and infinite in extent.
   b. There is radiation if the wire is curved, bent, discontinuous, terminated, or truncated, as shown in Figure 1.10.
3. If charge is oscillating in a time-motion, it radiates even if the wire is straight.

Curved



(a) Curved

Fig. 1.10a

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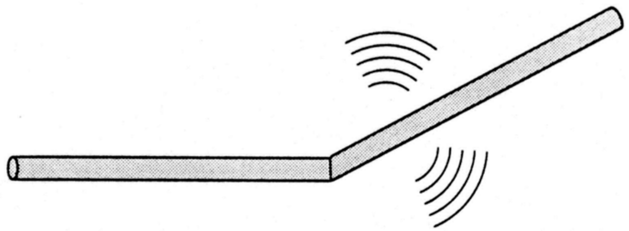
Bent



(b) Bent

Fig. 1.10b

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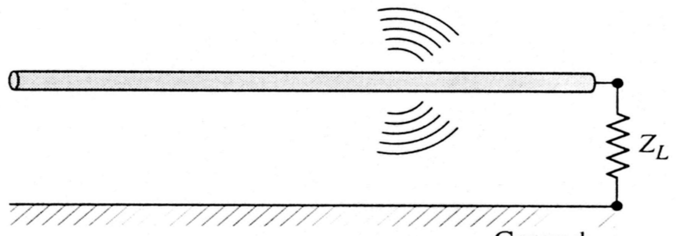
Terminated



(d) Terminated

Fig. 1.10d

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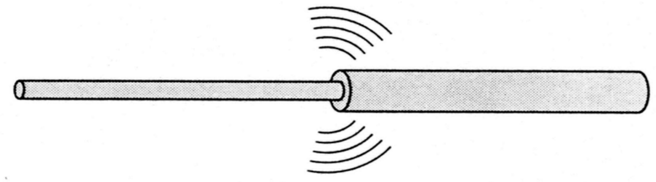
Discontinuous



(c) Discontinuous

Fig. 1.10c

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Pulses/Sinusoids

1. Pulses radiate a broad bandwidth (spectrum) of radiation. The shorter the pulse width, the broader the spectrum.
2. A sinusoidal (smooth) waveform of current or charge leads to a narrow spectrum of radiation; ideally zero bandwidth at the frequency of the sinusoid, if it continues indefinitely.

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Pulses In Wire

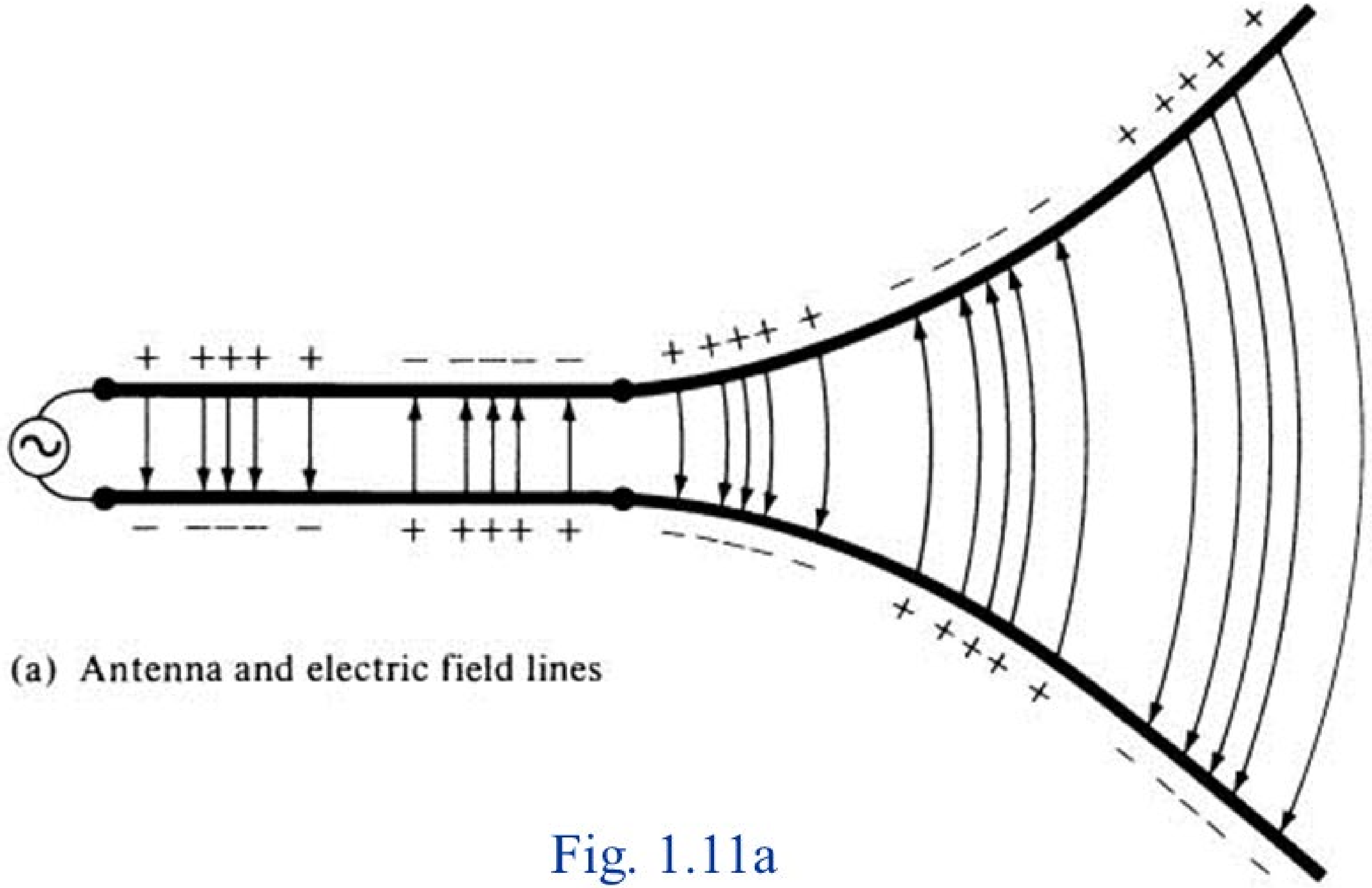
Stronger radiation with a more broad frequency spectrum occurs if the pulses are of shorter or more compact duration while continuous time-harmonic oscillating charge produces, ideally, radiation of single frequency determined by f oscillation.

Two Wires

Source, Line, Antenna, And Electric Field Lines

Chapter 1
Antennas

Antenna & Electric Field Lines



(a) Antenna and electric field lines

Fig. 1.11a

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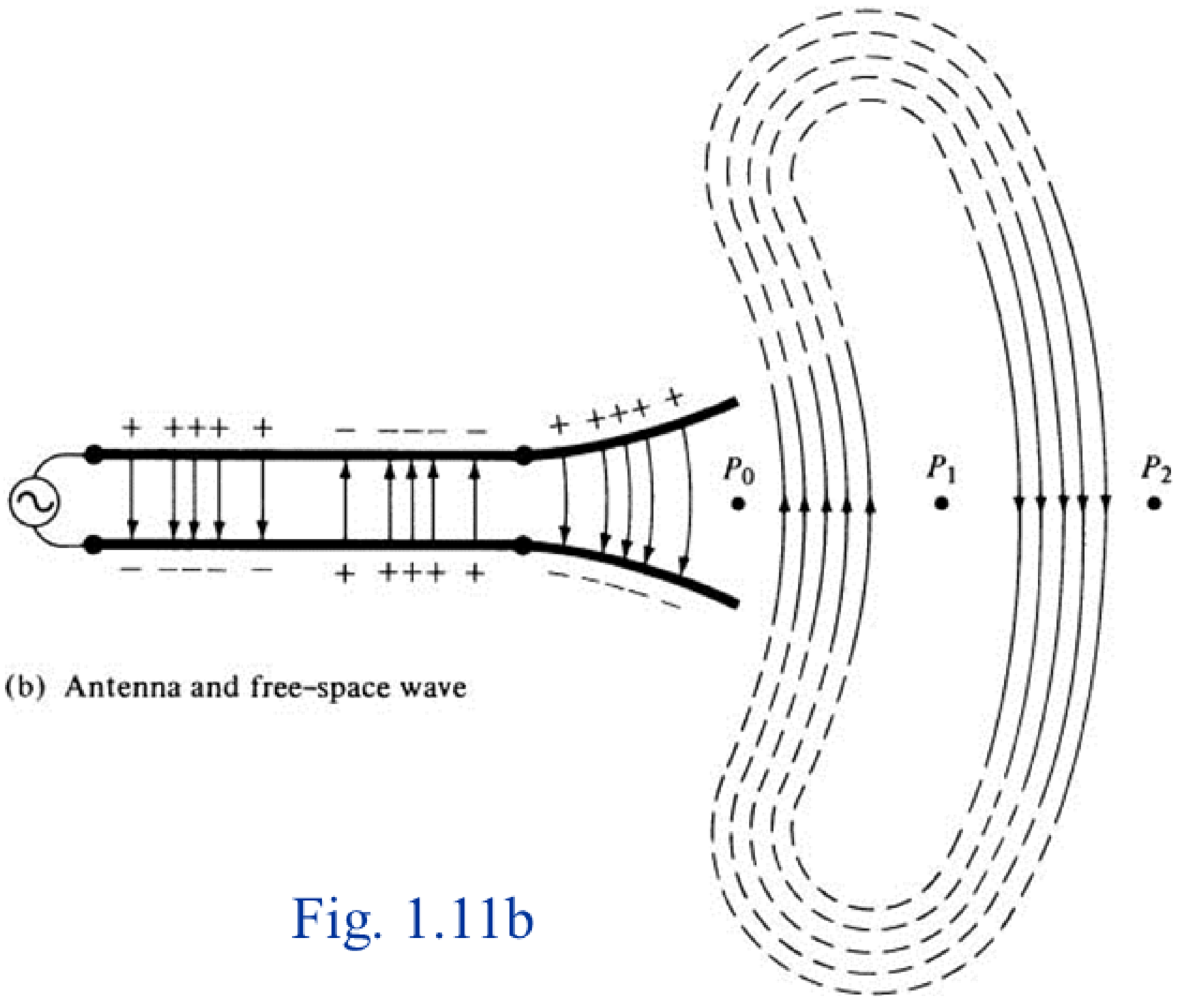
Antenna & Free-Space Wave



(b) Antenna and free-space wave

Fig. 1.11b

E-Field Lines of $\lambda/2$ Dipole

$t = 0$

$t = T/8$

$t = T/4$

$t = 3T/8$

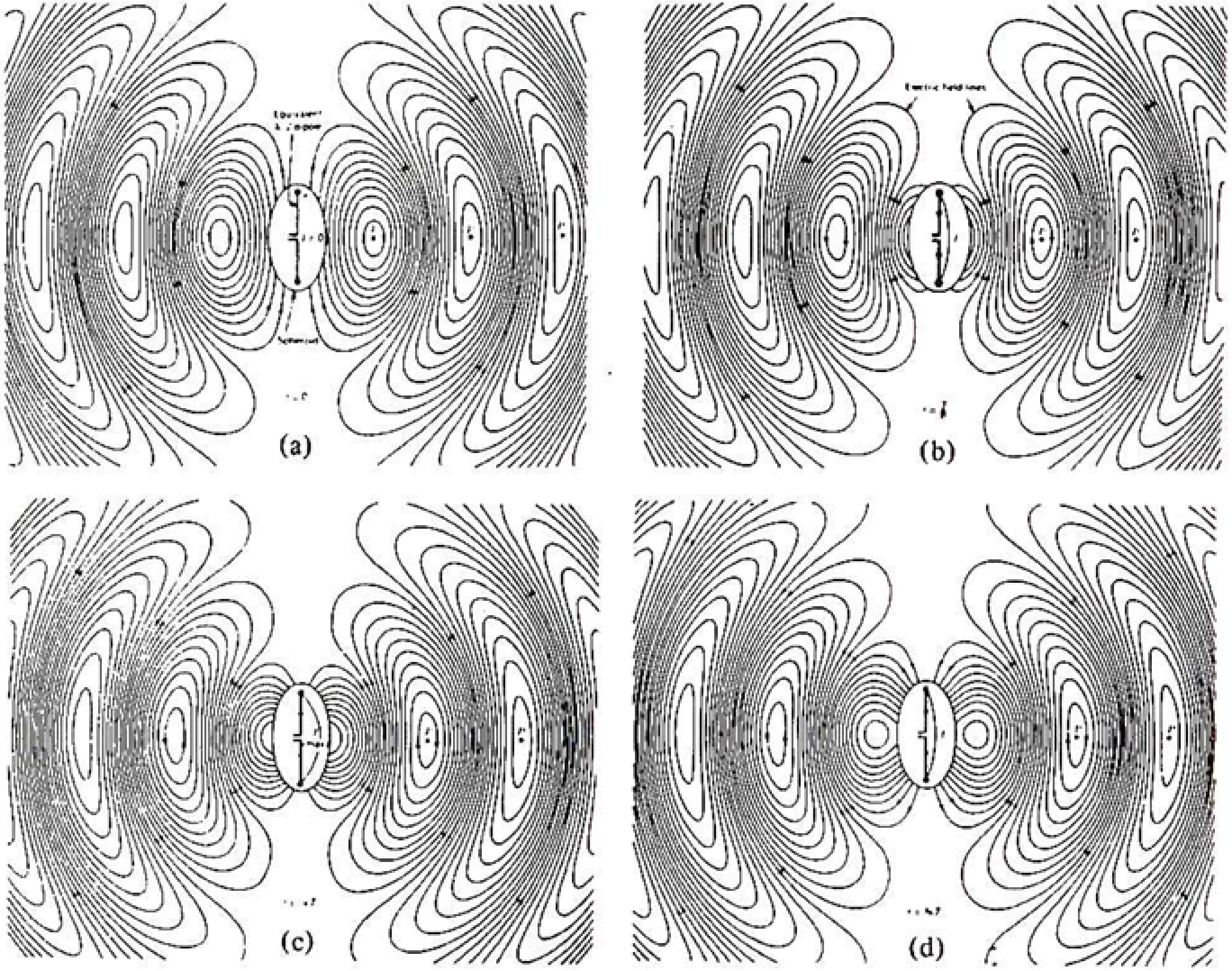
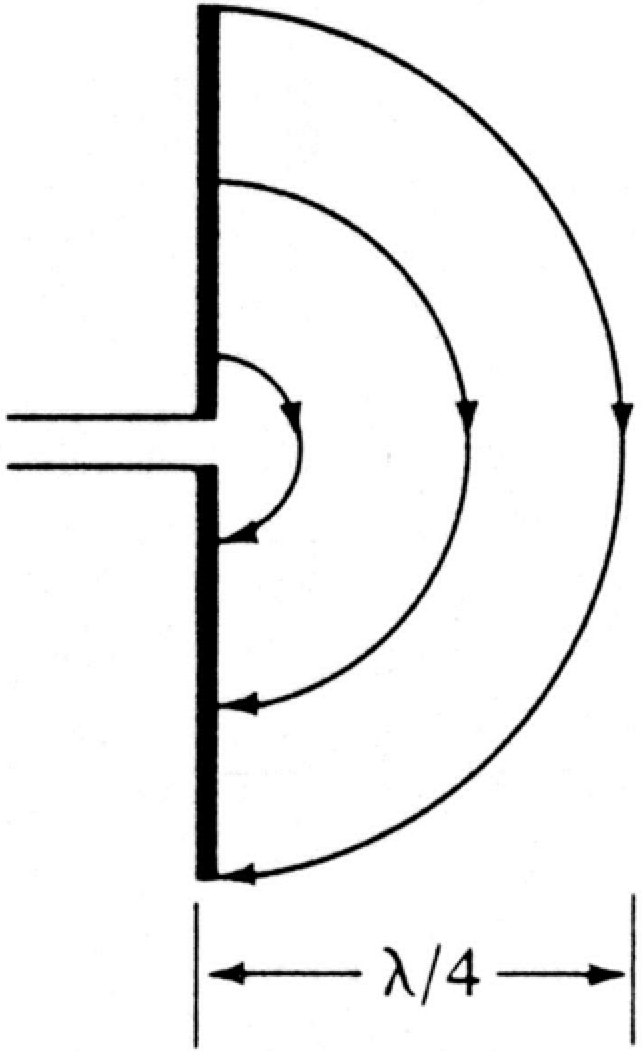


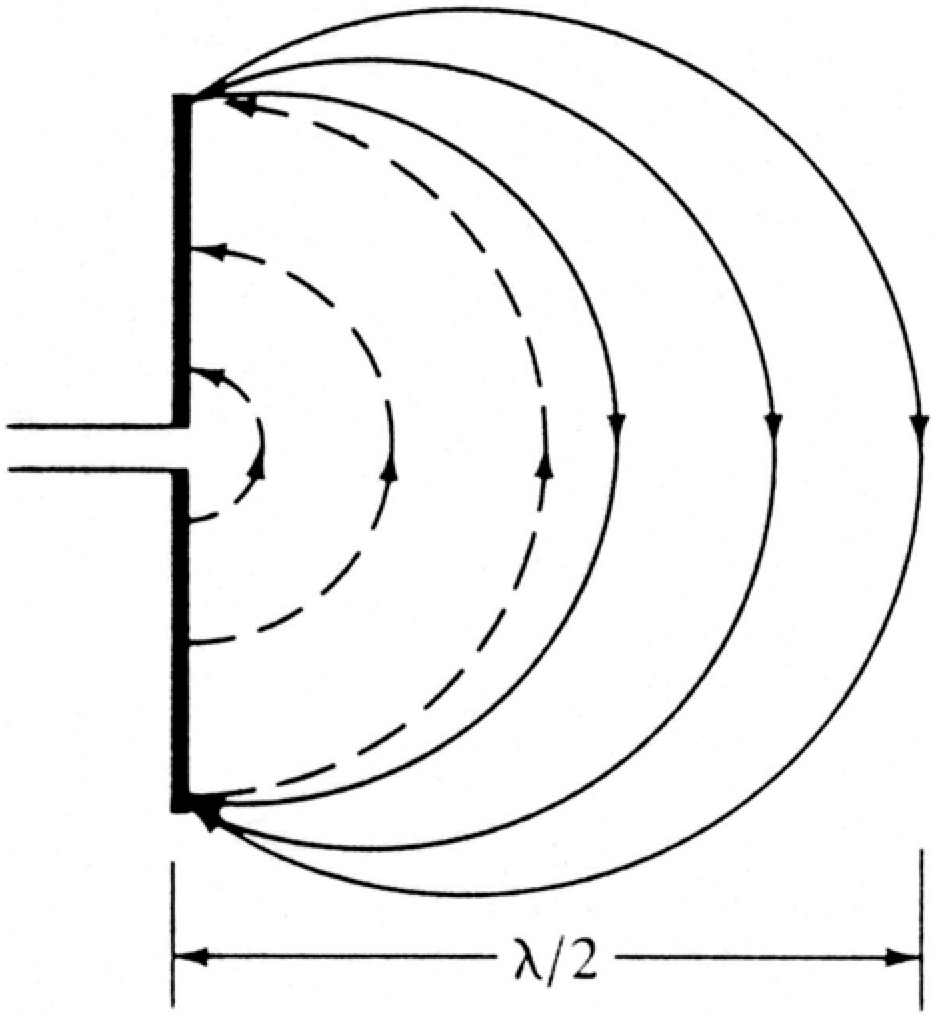
Fig. 1.12

Formation And Detachment of E-Field Lines For Short Dipole



(a) $t = T/4$ (T = period)

Fig. 1.14a



(b) $t = T/2$ (T = period)

Fig. 1.14b

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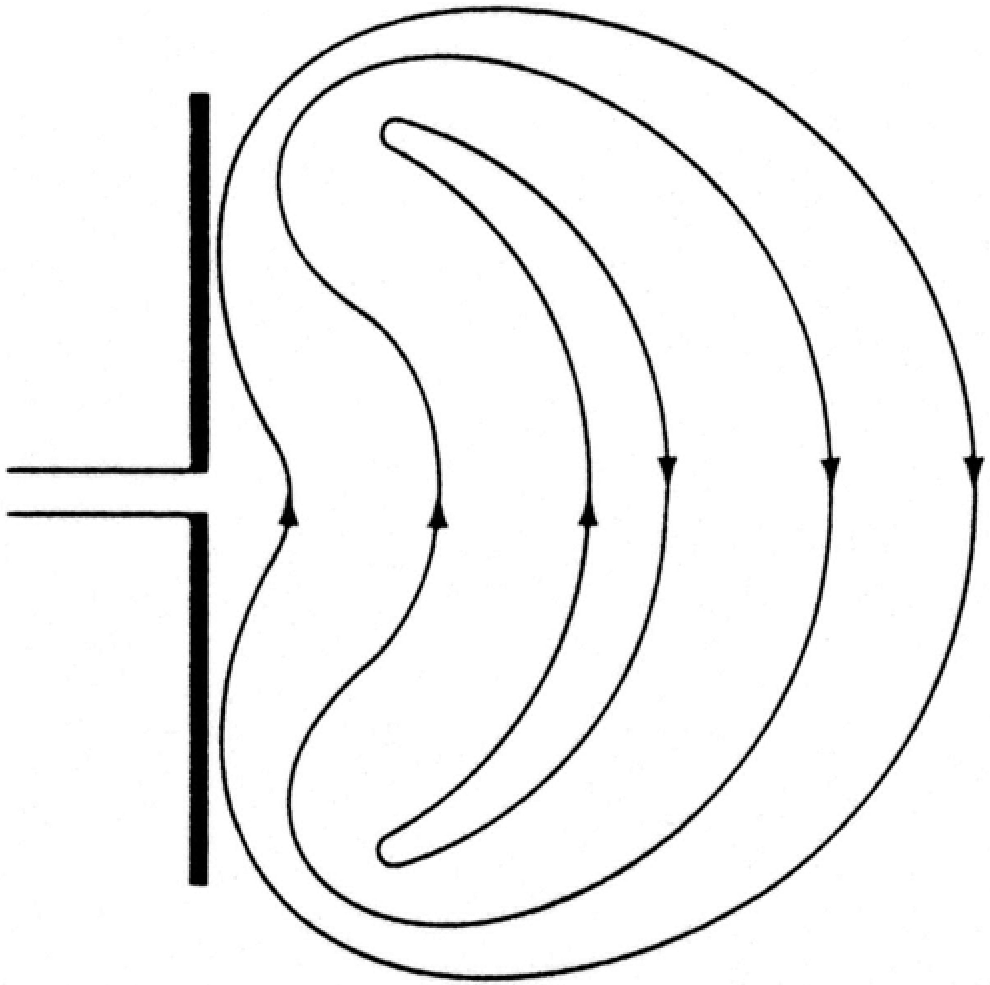

(b) $t = T/2$ (T = period)

Fig. 1.14c

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Antennas

Current Distribution of Transmission Line

And Linear Dipole

Two-Wire Transmission Line

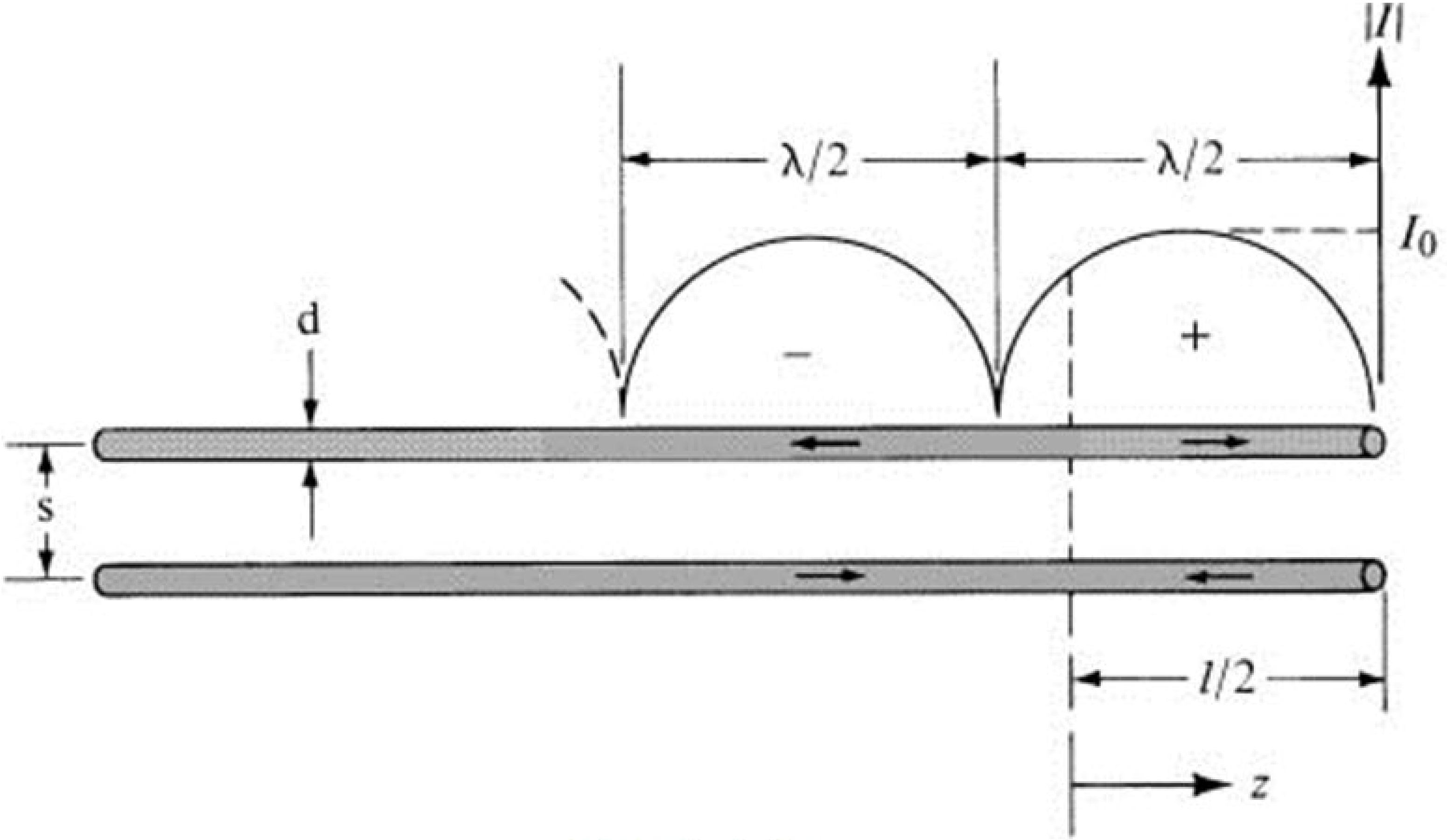


Fig. 1.15a

Flared-Transmission Line

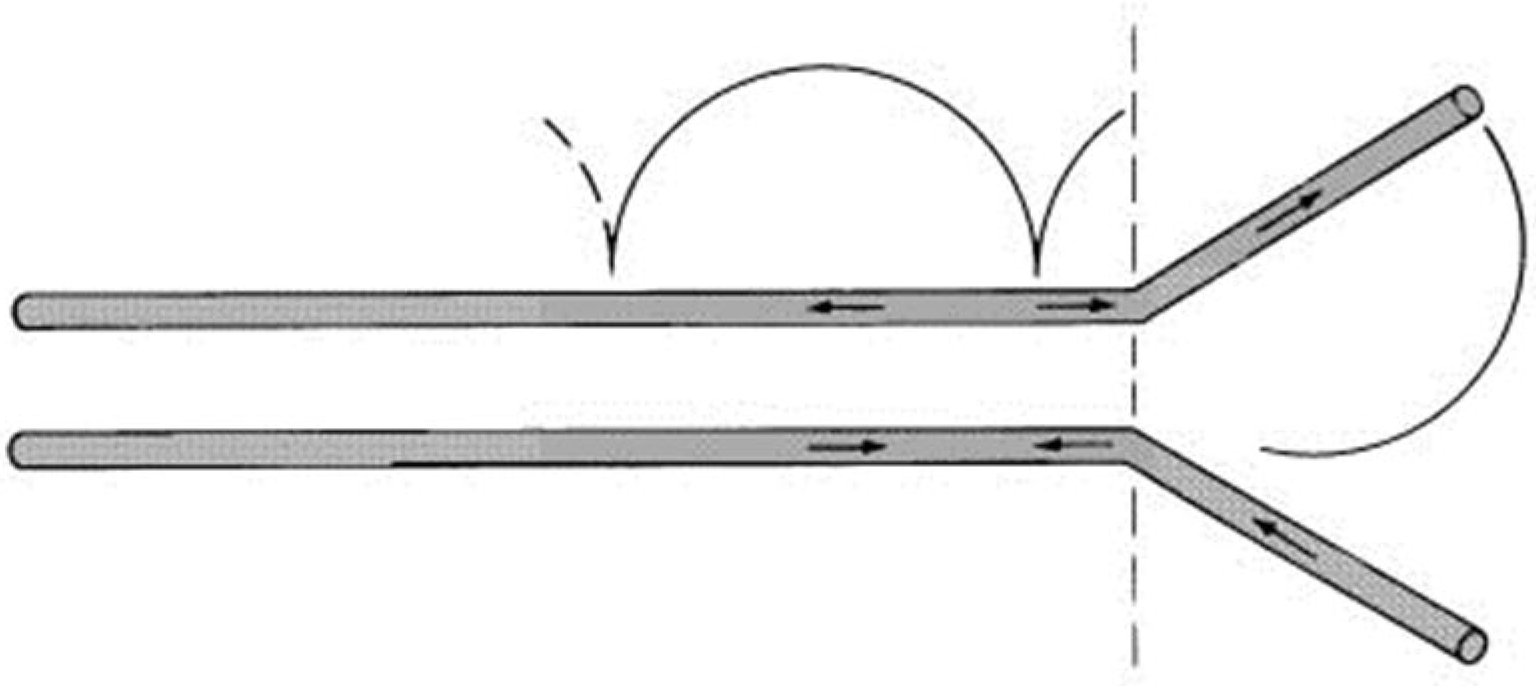


Fig. 1.15b

Linear Dipole

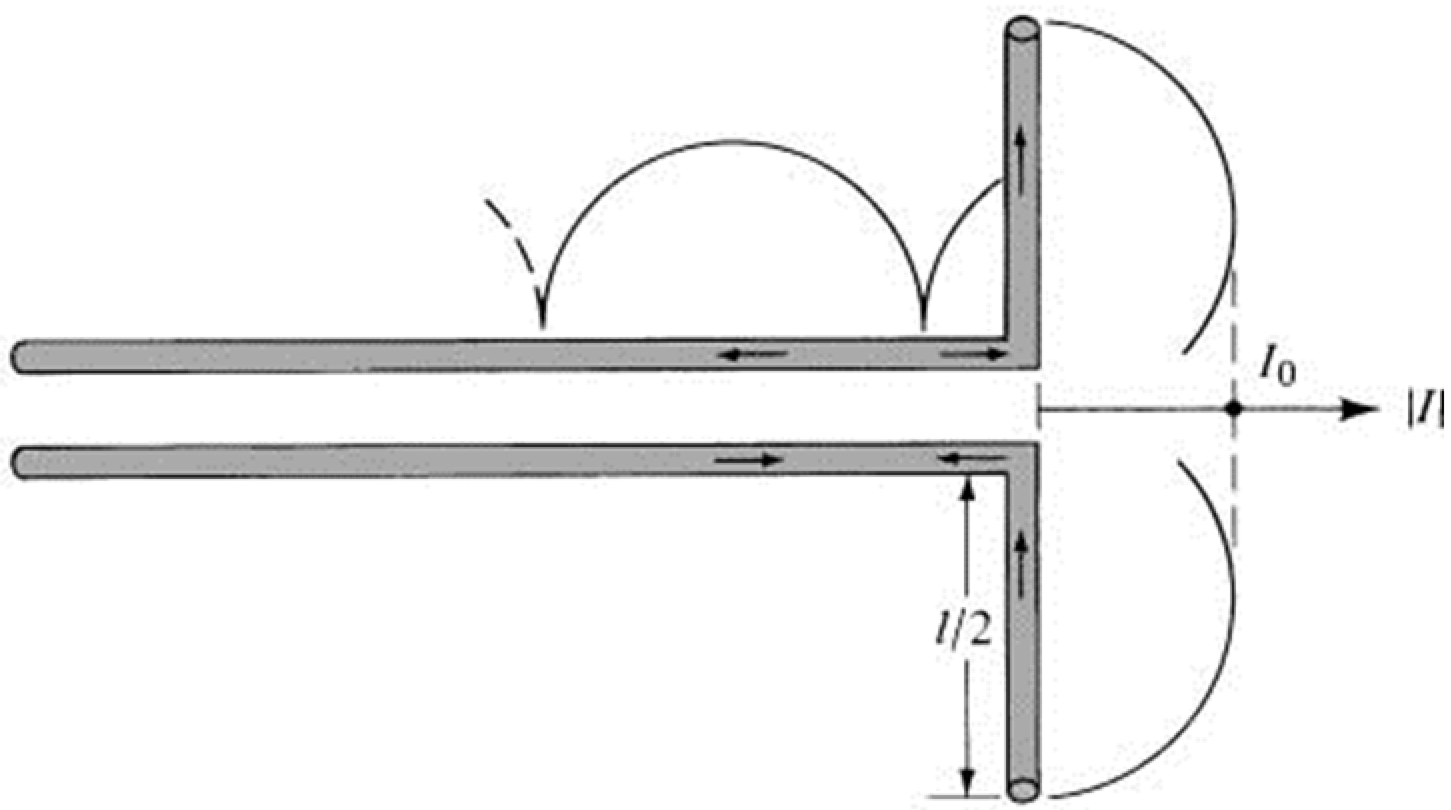
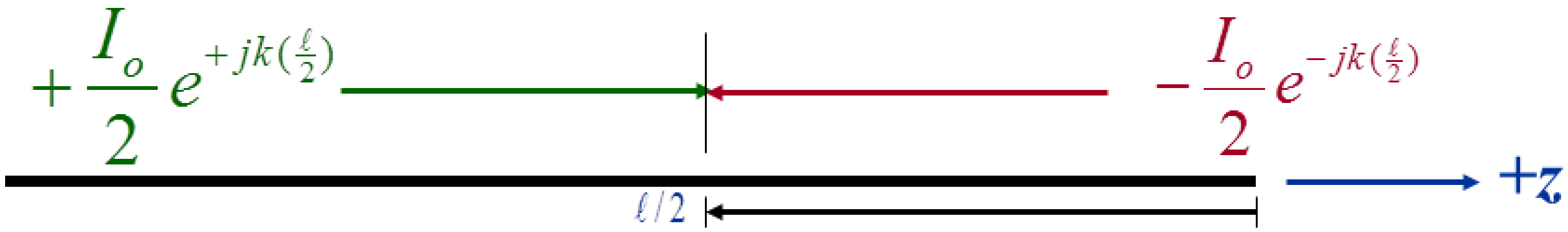


Fig. 1.15c

Formation of Standing Wave



$$I_t = +\frac{I_0}{2}e^{+jk\ell/2} - \frac{I_0}{2}e^{-jk\ell/2} = I_o j\left(\frac{e^{+jkl/2} - e^{-jk\ell/2}}{2j}\right)$$

$$I_t = jI_o \sin\left(\frac{k\ell}{2}\right) \Rightarrow |I_t| = |jI_o|\,|\sin\left(\frac{k\ell}{2}\right)|$$

$$\sin(x) = x - \frac{x^3}{3!} + \cdots \overset{x \to 0}{\approx} x$$

$$|I_t| = |jI_o|\,|\sin\left(\frac{k\ell}{2}\right)| \overset{k\ell/2 \to 0}{\approx} |I_o|\frac{k\ell}{2}$$

Current Distribution Linear Dipole

$\ell << \lambda$

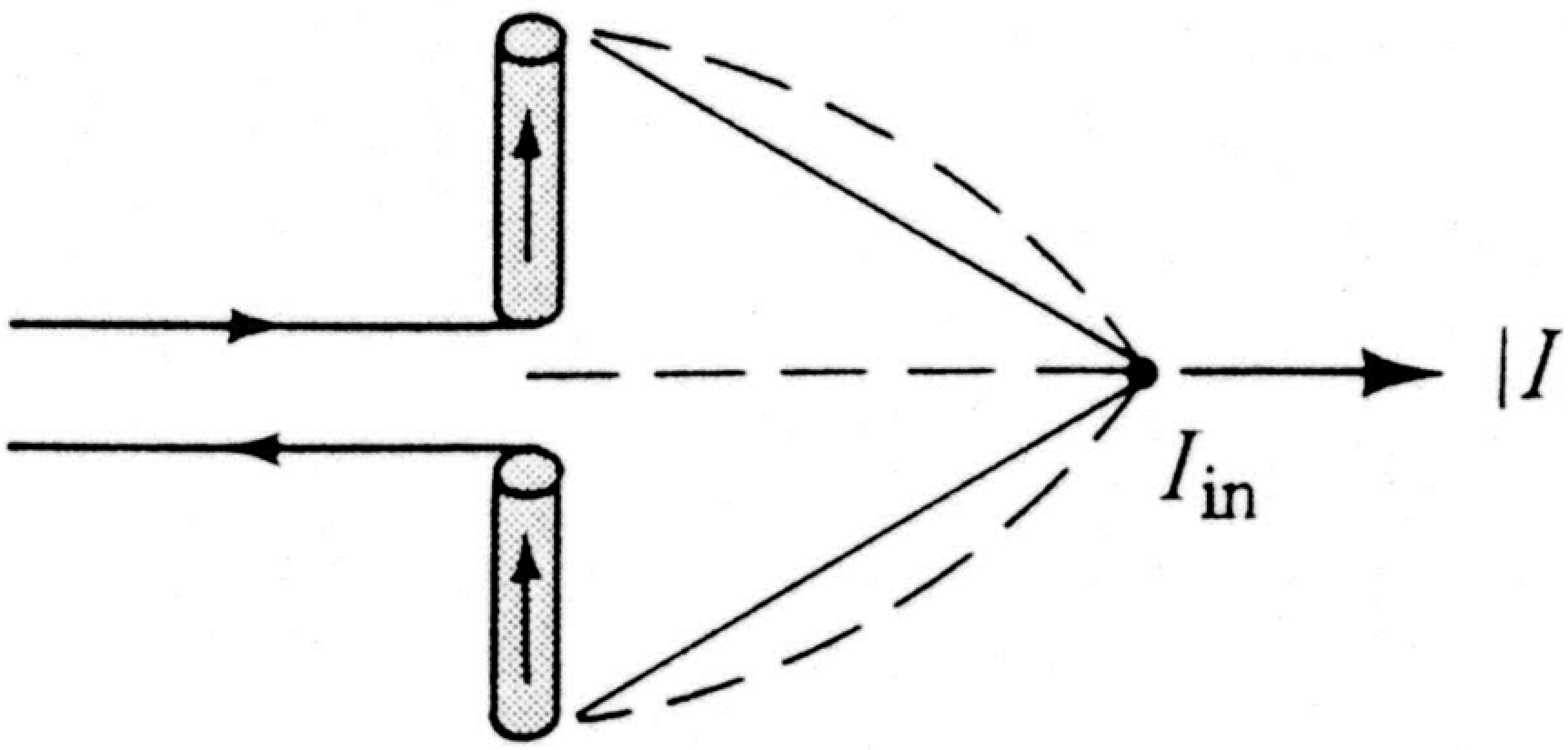


Fig. 1.16a

$$\ell = \lambda/2$$

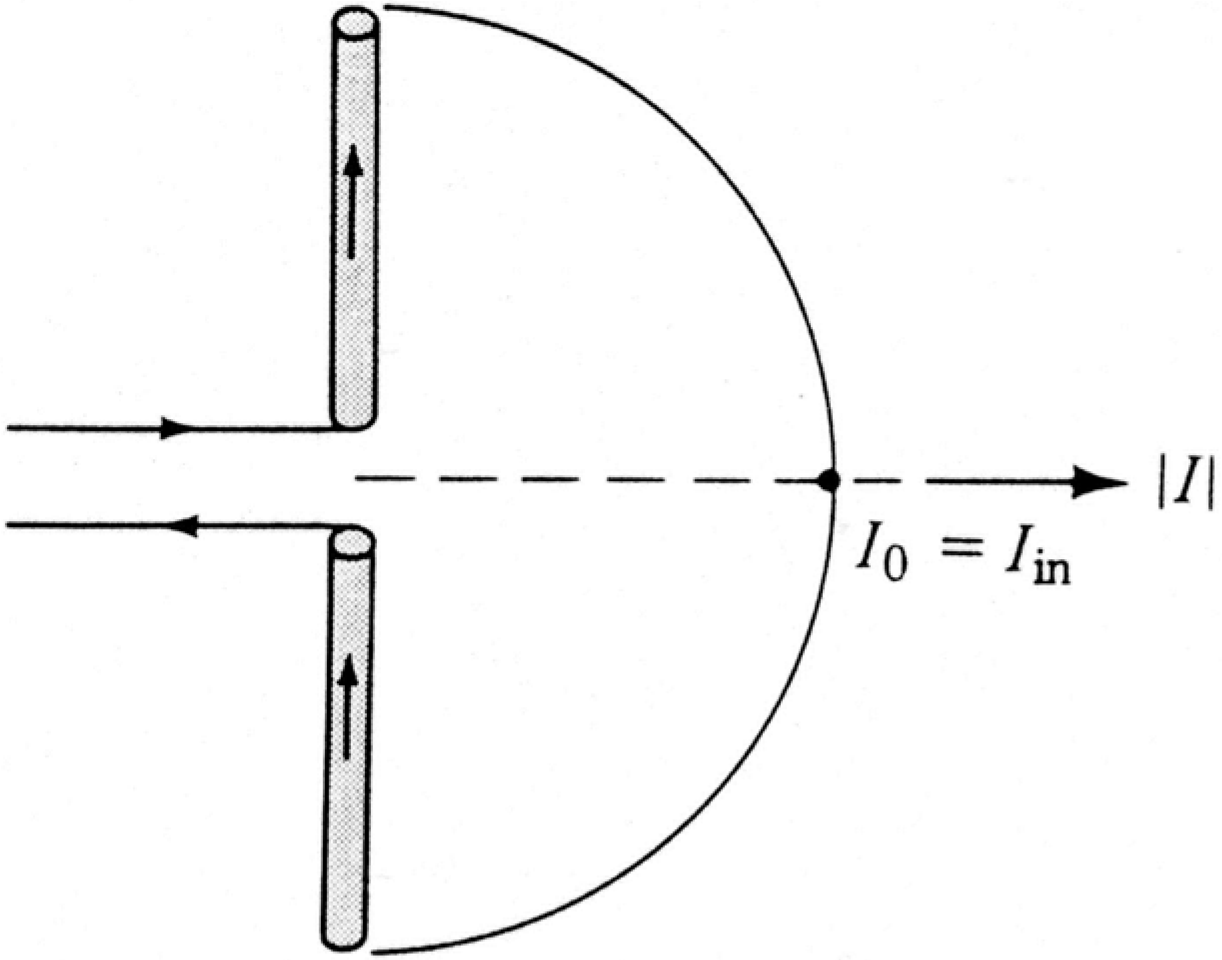


Fig. 1.16b

$\lambda/2 < \ell < \lambda$

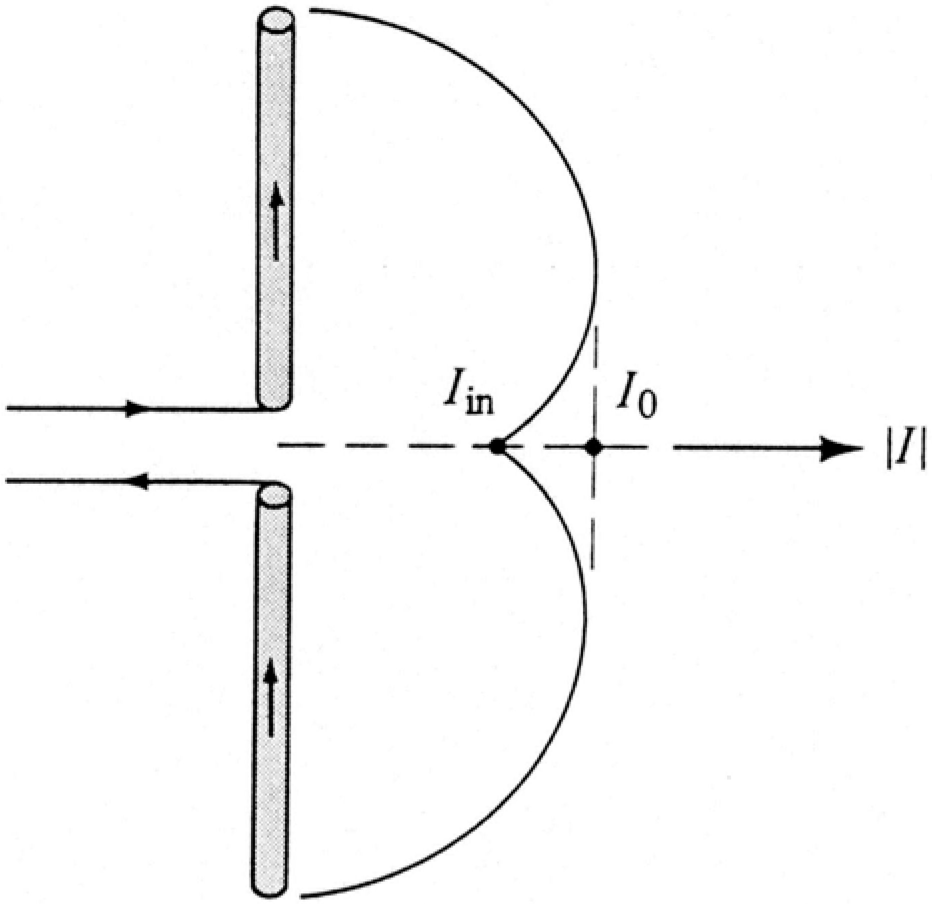


Fig. 1.16c

$\lambda < \ell < 3\lambda/2$

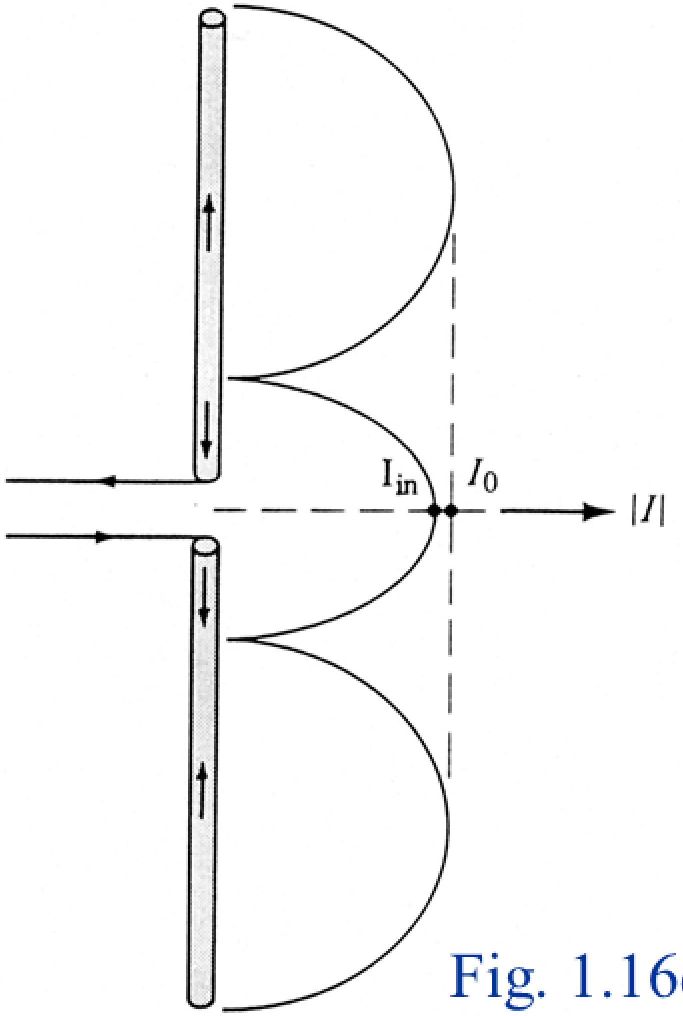


Fig. 1.16d

Current Variation As A Function of Time For $\lambda/2$ Dipole

$\ell = \lambda/2$

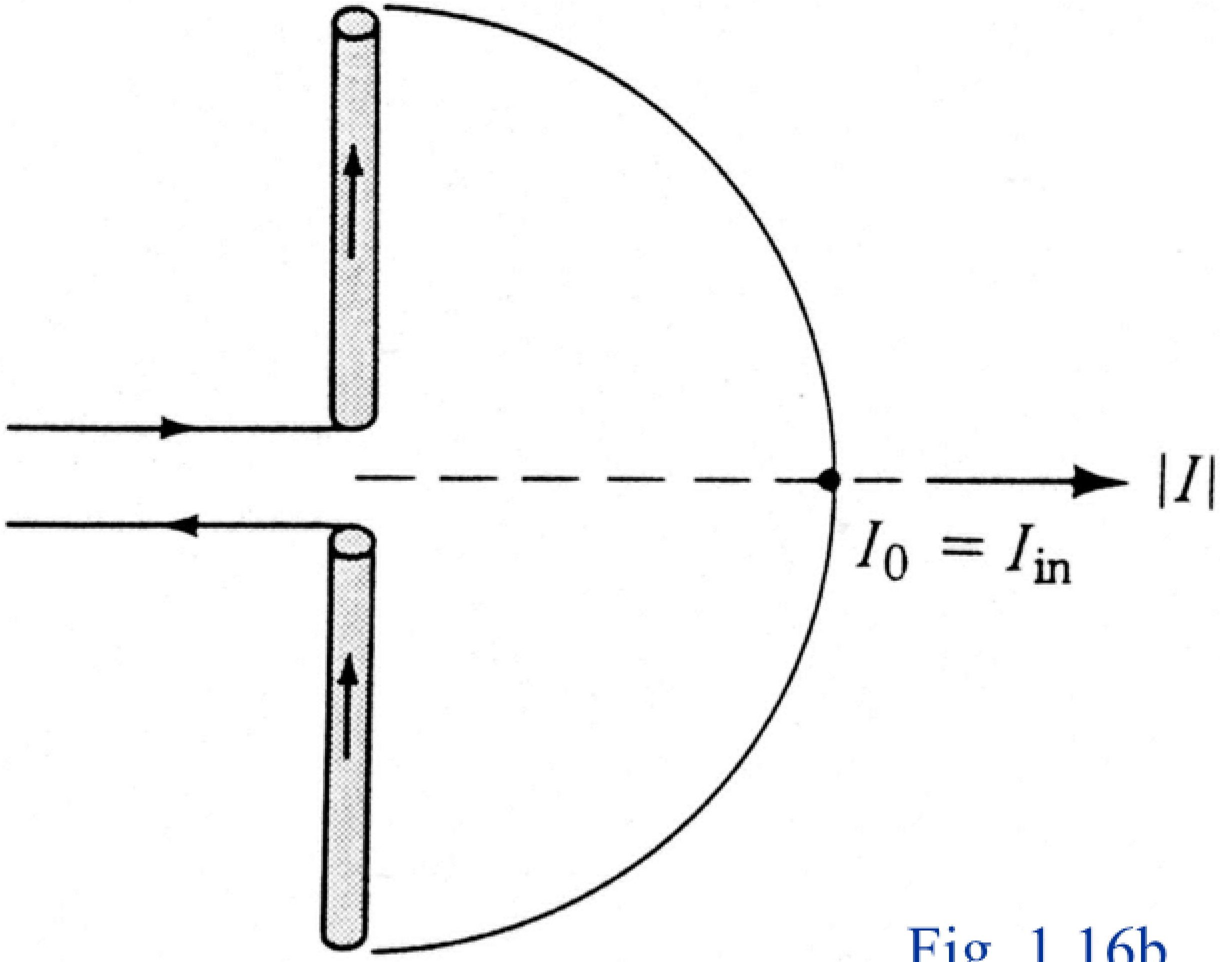


Fig. 1.16b

$t = 0$

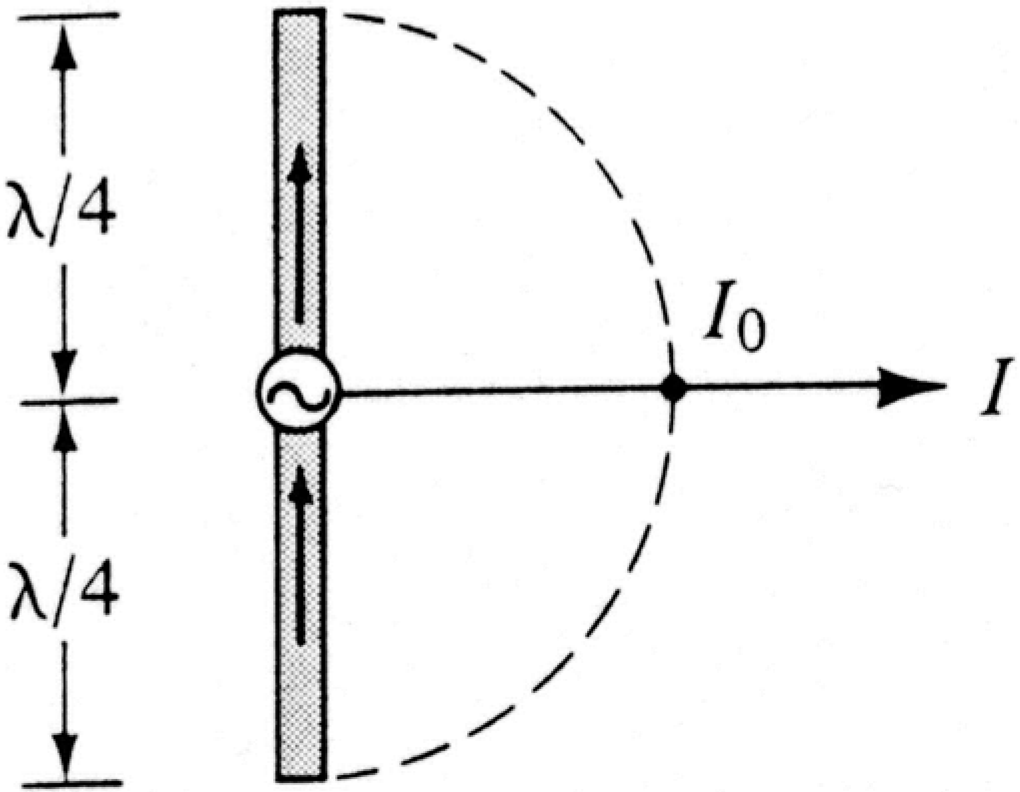


Fig. 1.17a

$$t = T/8$$

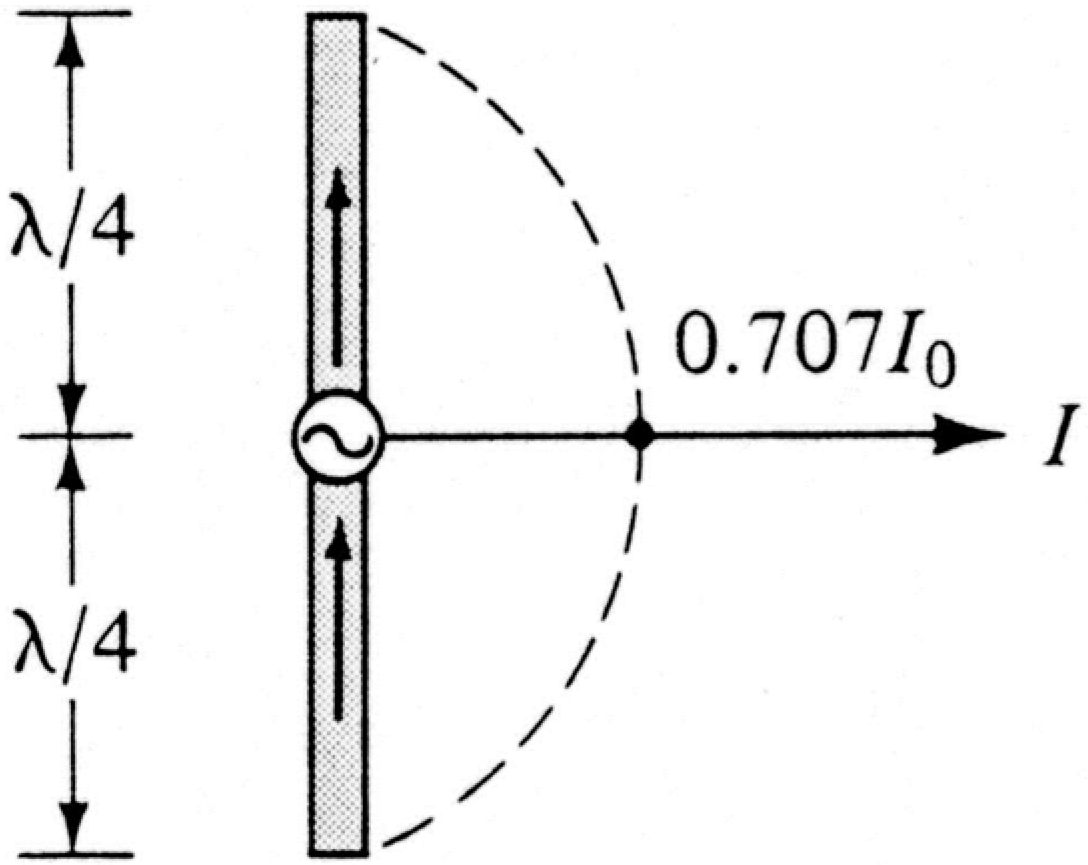


Fig. 1.17b

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$t = T/4$

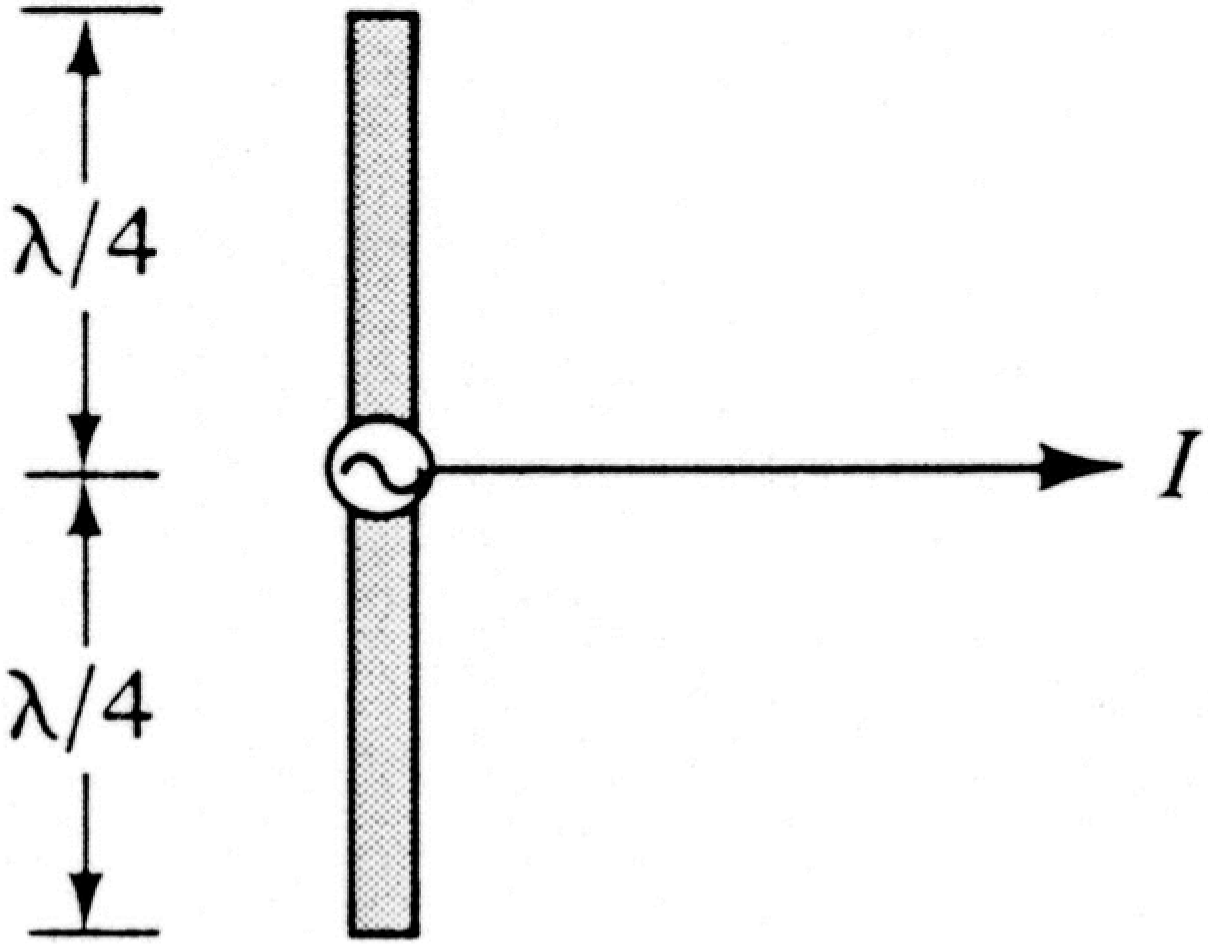

Fig. 1.17c

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$t = 3T/8$

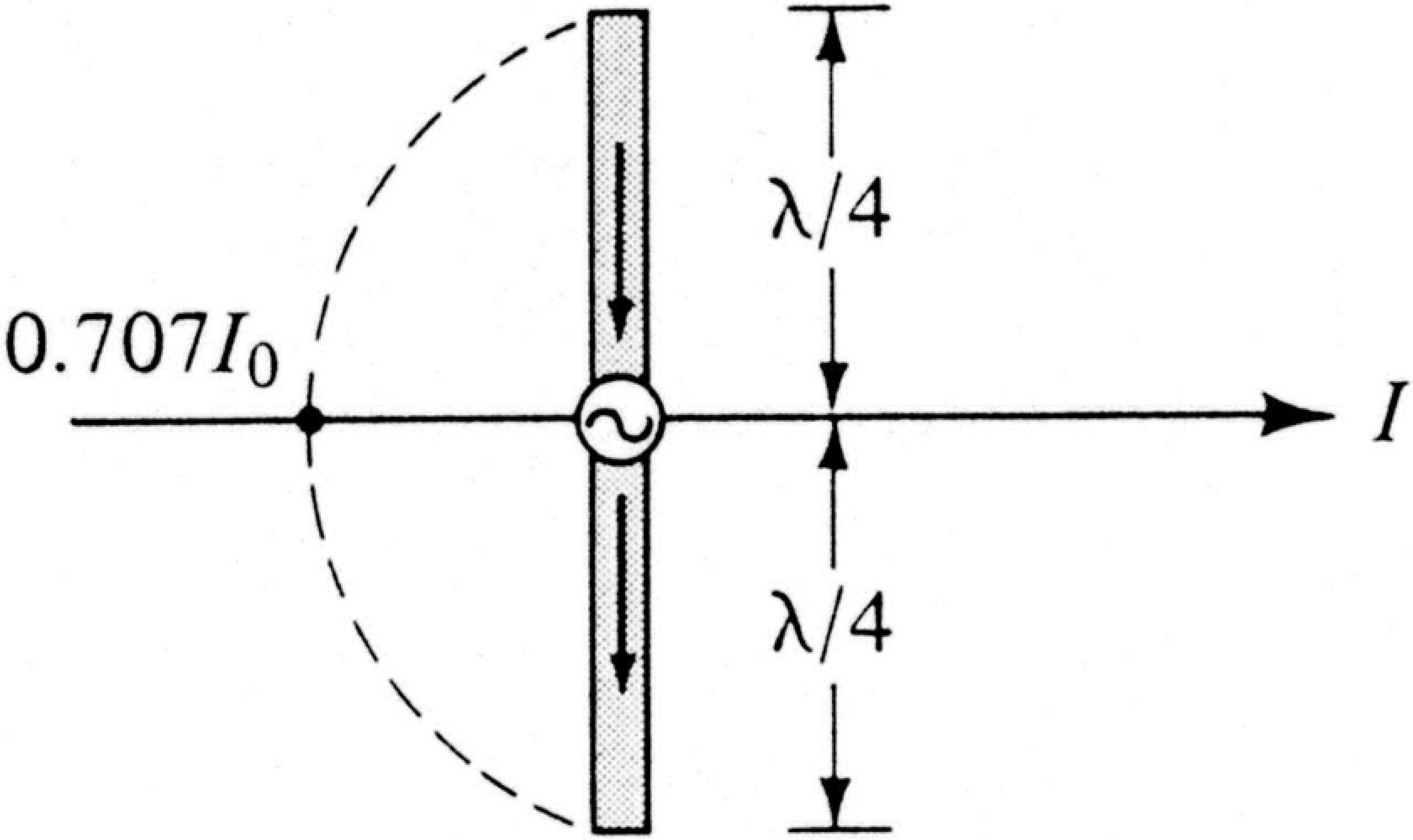

Fig. 1.17d

$$t = T/2$$

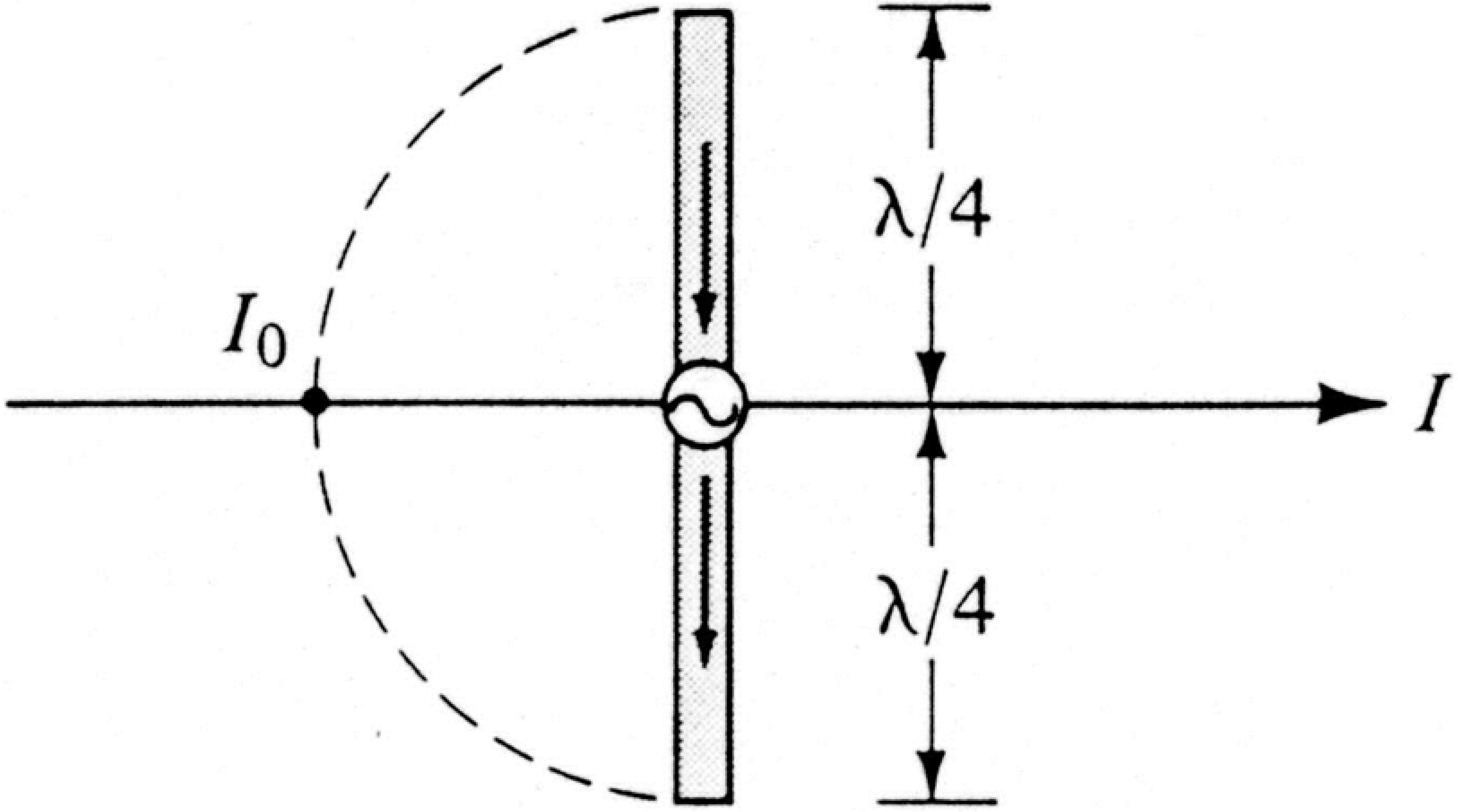

Fig. 1.17e